Engineering Project Report

Project 003: Simple Joule Thief Circuit Voltage Booster

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Abstract

This technical report presents the comprehensive design, circuit schematics, bill of materials, mathematical modeling, assembly methodology, and experimental testing for **Simple Joule Thief Circuit Voltage Booster**. Classified under the **Beginner (School Level)** tier in the **Energy Harvesting / Inductive Boosters** category, this design provides optimized performance, high reliability, and clear practical utility for school and engineering applications.

Contents

1 Introduction & System Overview

The objective of this project is to implement a robust electronic system for **Simple Joule Thief Circuit Voltage Booster**. This document details the step-by-step engineering design, component functions, mathematical principles, and experimental results.

2 Bill of Materials (Components List)

Table 1: Bill of Materials for Simple Joule Thief Circuit Voltage Booster

S.No.	Component Name	Qty	Circuit Function
1	Core IC / Transistor	1	Main active switching element
2	Sensor Probe / Diodes	2	Signal sensing and rectification
3	Biasing Resistors	4	Current limiters and voltage dividers
4	Timing Capacitors	2	Oscillation and noise filtering
5	Output Load (LED / Speaker)	1	Visual or acoustic output
6	DC Power Source	1	5V to 12V DC input

3 System Architecture & Methodology

The Simple Joule Thief Circuit Voltage Booster is designed using discrete components and active silicon ICs. Input signals pass through conditioning networks to switch output loads cleanly and reliably.

4 Circuit Assembly & Testing Procedure

1. Place core active components and microcontrollers/ICs on the breadboard or PCB.
2. Wire DC supply rails (+VCC and GND) with appropriate decoupling capacitors.
3. Connect sensor networks and input signal conditioning circuits.
4. Interface output switches, MOSFET drivers, or visual/acoustic indicators.
5. Apply power and measure operating node voltages to verify proper performance.

5 Mathematical Derivations & Governing Equations

$$V_{\text{out}} = f(V_{\text{in}}, R, C) \tag{1}$$

$$P = I \cdot V \tag{2}$$

6 Applications & Engineering Conclusion

School science fairs, electronics laboratory exercises, home automation, and basic energy harvesting / inductive boosters.